Model 2 — Data-Driven Linear Model (DMDc) and Optimal Control (LQR / MPC)

Learn A, B from flight data, design the controller on the learned model, estimate the state with an EKF, fly the nonlinear drone

1 Nonlinear Flight Data — DRONE / SIMULATION

$$\boldsymbol{x}_k = \begin{bmatrix} \boldsymbol{p}_k - \boldsymbol{p}_{\text{ref}} \\ \boldsymbol{v}_k \\ \text{Log}(R_k) \\ \boldsymbol{\omega}_k \end{bmatrix} \in \mathbb{R}^{12}, \qquad \boldsymbol{u}_k = \begin{bmatrix} T_k - mg_0 \\ \boldsymbol{\tau}_k \end{bmatrix} \in \mathbb{R}^4$$

sampled at $t_k = k\,\Delta t, \quad k = 1, \ldots, N$

IN flights with a stabilising pilot + random excitation
OUT state history $\{\boldsymbol{x}_k\}$, input history $\{\boldsymbol{u}_k\}$

$\{\boldsymbol{x}_k\}, \{\boldsymbol{u}_k\}$

2 DMDc — SYSTEM IDENTIFICATION

$$X = [\boldsymbol{x}_1 \cdots \boldsymbol{x}_{N-1}], \qquad X^+ = [\boldsymbol{x}_2 \cdots \boldsymbol{x}_N]$$
$$U = [\boldsymbol{u}_1 \cdots \boldsymbol{u}_{N-1}]$$
$$[\,A \;\; B\,] = X^+ \begin{bmatrix} X \\ U \end{bmatrix}^{\dagger} \;\Rightarrow\; \boldsymbol{x}_{k+1} \approx A\boldsymbol{x}_k + B\boldsymbol{u}_k$$

IN $X, X^+ \in \mathbb{R}^{12\times(N-1)}$, $U \in \mathbb{R}^{4\times(N-1)}$
OUT $A \in \mathbb{R}^{12\times12}$, $B \in \mathbb{R}^{12\times4}$ (discrete time)

A, B

LQR BRANCH (infinite horizon)

3 Bellman Equation — OPTIMAL CONTROL

$$J = \sum_{k=0}^{\infty} \left(\boldsymbol{x}_k^\top Q\, \boldsymbol{x}_k + \boldsymbol{u}_k^\top R\, \boldsymbol{u}_k \right)$$
$$V(\boldsymbol{x}_k) = \min_{\boldsymbol{u}_k} \left[\boldsymbol{x}_k^\top Q \boldsymbol{x}_k + \boldsymbol{u}_k^\top R \boldsymbol{u}_k + V(\boldsymbol{x}_{k+1}) \right]$$
$$V(\boldsymbol{x}) = \boldsymbol{x}^\top P\, \boldsymbol{x} \quad \text{(quadratic value function)}$$

IN A, B; weights $Q \succeq 0$, $R \succ 0$
OUT optimal input is linear: $\boldsymbol{u}^* = -K\boldsymbol{x}$

V quadratic $\Rightarrow$ $\boldsymbol{u}^*$ linear in $\boldsymbol{x}$

4 Linear Quadratic Regulator — OPTIMAL CONTROL

$$K = \left(R + B^\top P B \right)^{-1} B^\top P A$$
$$\boldsymbol{u}_k = -K\, \boldsymbol{x}_k$$

IN A, B, R and value matrix P
OUT feedback gain $K \in \mathbb{R}^{4\times12}$

needs P

5 Riccati Recursion — ITERATIVE

$$P_{k+1} = A^\top P_k A - A^\top P_k B \left(R + B^\top P_k B \right)^{-1} B^\top P_k A + Q$$

IN A, B, Q, R; start $P_0 = Q$
OUT sequence $P_0, P_1, \cdots \to P_\infty$

$P_k \to P_\infty$ $(k \to \infty)$

6 DARE — STEADY STATE

$$P = A^\top P A - A^\top P B \left(R + B^\top P B \right)^{-1} B^\top P A + Q$$

IN A, B, Q, R
OUT $P \in \mathbb{R}^{12\times12}$, $P \succ 0 \Rightarrow$ final K

$\boldsymbol{u}_k = -K\hat{\boldsymbol{x}}_k$

logged $\hat{\boldsymbol{x}}_k, \boldsymbol{u}_k \Rightarrow$ re-identify A, B

converged P

$P_f = P$ (terminal cost of the MPC)

MPC BRANCH (finite horizon + constraints)

7 Model Predictive Control — CONSTRAINED

$$\min_{\boldsymbol{u}_0, \ldots, \boldsymbol{u}_{N-1}} \sum_{k=0}^{N-1} \left(\boldsymbol{x}_k^\top Q \boldsymbol{x}_k + \boldsymbol{u}_k^\top R \boldsymbol{u}_k \right) + \boldsymbol{x}_N^\top P_f\, \boldsymbol{x}_N$$
$$\text{s.t.} \quad \boldsymbol{x}_{k+1} = A\boldsymbol{x}_k + B\boldsymbol{u}_k, \quad \boldsymbol{u}_k \in \mathcal{U}, \quad \boldsymbol{x}_k \in \mathcal{X}$$

IN A, B, Q, R, N, limits $\mathcal{U}, \mathcal{X}$, P_f; start $\boldsymbol{x}_0 = \hat{\boldsymbol{x}}_k$ (block 11)
OUT finite-horizon optimal control problem

stack over $k = 0..N-1$

8 Quadratic Program — STACKED FORM

$$\min_{\mathbf{U}} \; \tfrac{1}{2}\mathbf{U}^\top H \mathbf{U} + \boldsymbol{f}^\top \mathbf{U} \quad \text{s.t.} \quad G\mathbf{U} \le \boldsymbol{h}, \quad A_{eq}\mathbf{U} = \boldsymbol{b}_{eq}$$
$$H = 2\left(S_u^\top \bar{Q} S_u + \bar{R} \right), \qquad \boldsymbol{f} = 2 S_u^\top \bar{Q} S_x \boldsymbol{x}_0$$

IN MPC problem, current estimate $\boldsymbol{x}_0 = \hat{\boldsymbol{x}}_k$
OUT $H \in \mathbb{R}^{4N\times4N}$, $\boldsymbol{f}$, G, $\boldsymbol{h} \to$ QP solver

solve QP $\Rightarrow$ $\mathbf{U}^*$

9 MPC Control Input — RECEDING HORIZON

$$\mathbf{U}^* = [\,\boldsymbol{u}_0^*, \boldsymbol{u}_1^*, \ldots, \boldsymbol{u}_{N-1}^*\,] \longrightarrow$$
apply only $\boldsymbol{u}_0^*$, then re-solve at $k+1$

IN QP data $(H, \boldsymbol{f}, G, \boldsymbol{h})$ at time k
OUT $\boldsymbol{u}_0^* \in \mathbb{R}^4$

$\boldsymbol{u}_0^*$

estimate $\hat{\boldsymbol{x}}_k$ = MPC start $\boldsymbol{x}_0$ (every Δt)

10 Apply to Drone — ACTUATION

$$\boldsymbol{u}_k = \boldsymbol{u}_{\text{trim}} + \begin{cases} \boldsymbol{u}_0^*(\hat{\boldsymbol{x}}_k) & \text{(MPC)} \\ -K\hat{\boldsymbol{x}}_k & \text{(LQR)} \end{cases} = \begin{bmatrix} T_k \\ \boldsymbol{\tau}_k \end{bmatrix} \longrightarrow \boldsymbol{f} = \Gamma^{-1}\boldsymbol{u}_k \longrightarrow \Omega_i = \sqrt{f_i / k_f}$$

IN estimate $\hat{\boldsymbol{x}}_k$ (block 11), controller output
OUT motor speeds $\boldsymbol{\Omega} \in \mathbb{R}^4 \to$ the drone moves to $\boldsymbol{x}_{k+1}$

$\hat{\boldsymbol{x}}_k$ for LQR

drone moves $\Rightarrow$ camera pixels $\boldsymbol{z}_j$, gyro $\boldsymbol{z}_\omega$, applied $\boldsymbol{u}_k$

11 State Estimation: Extended Kalman Filter — ESTIMATION (VIO)

Prediction with the learned model $(F = A)$

$$\hat{\boldsymbol{x}}_{k+1} = A\,\hat{\boldsymbol{x}}_k + B\,\boldsymbol{u}_k$$
$$\Sigma_{k+1} = A\,\Sigma_k A^\top + Q_e$$
$$Q_e = \text{cov}\left(\boldsymbol{x}_{k+1} - A\boldsymbol{x}_k - B\boldsymbol{u}_k \right)$$

Correction with camera pixels and gyroscope

$$\boldsymbol{q}_j = R_{bc}^\top R(\boldsymbol{\theta})^\top (\boldsymbol{\ell}_j - \boldsymbol{p}), \quad \boldsymbol{z}_j = \begin{bmatrix} f\,q_1/q_3 + c_x \\ f\,q_2/q_3 + c_y \end{bmatrix}, \quad \boldsymbol{z}_\omega = \boldsymbol{\omega}$$
$$H = \partial h / \partial \boldsymbol{x}, \qquad L = \Sigma H^\top \left(H \Sigma H^\top + R_n \right)^{-1}$$
$$\hat{\boldsymbol{x}} \leftarrow \hat{\boldsymbol{x}} + L\left(\boldsymbol{z} - h(\hat{\boldsymbol{x}}) \right), \quad \Sigma \leftarrow (I - LH)\,\Sigma\,(I - LH)^\top + L R_n L^\top$$

IN applied $\boldsymbol{u}_k$; pixels $\boldsymbol{z}_j$ of known floor markers and gyro $\boldsymbol{z}_\omega$ (25 Hz); learned A, B
OUT estimate $\hat{\boldsymbol{x}}_k \in \mathbb{R}^{12}$ and its covariance $\Sigma_k \in \mathbb{R}^{12\times12}$ $\to$ blocks 9 and 10

signal / result passed on — matrix re-used by another block — feedback: the EKF estimate returns to the controllers

When does each block run? **Offline, once:** 1 (record) $\to$ 2 (DMDc, and Q_e for block 11) $\to$ 3–6 (P, K).
Online, every $\Delta t = 40$ **ms:** 11 (estimate) $\to$ LQR: 10; or 11 $\to$ MPC: 8 $\to$ 9 $\to$ 10.
Toy-model settings: $N = 25$ (1 s horizon), $|v_i| \le 1$ m/s (`max_velocity` in controller.yaml), $|\phi|, |\theta| \le 20°$, $5\,\text{N} \le T \le 25\,\text{N}$; camera 640×480 px, 121 floor markers, 1 px noise.

Notation and dimensions

Data and learned model (blocks 1–2)

$\boldsymbol{x}_k$	state deviation from hover	$\mathbb{R}^{12}$	m, m/s, rad, rad/s
$\boldsymbol{u}_k$	input deviation $[T_k - mg_0; \boldsymbol{\tau}_k]$	$\mathbb{R}^4$	N, N m
$\boldsymbol{p}_k, \boldsymbol{v}_k, \boldsymbol{\omega}_k$	position, velocity, body rate	$\mathbb{R}^3$	m, m/s, rad/s
R_k, $\text{Log}(R_k)$	attitude, its rotation vector	$\text{SO}(3)$, $\mathbb{R}^3$	–, rad
$\boldsymbol{p}_{\text{ref}}$	hover point $[0, 0, 2]^\top$	$\mathbb{R}^3$	m
mg_0	hover thrust (1.5×9.81)	$\mathbb{R}$	N (14.715)
Δt, t_k	sample time, $t_k = k\Delta t$	$\mathbb{R}$	s (0.04)
N (blocks 1–2)	number of recorded samples	$\mathbb{N}$	–
X, X^+	snapshots, shifted by one sample	$\mathbb{R}^{12\times(N-1)}$	mixed
U	input snapshot matrix	$\mathbb{R}^{4\times(N-1)}$	N, N m
$[X; U]^\dagger$	Moore–Penrose pseudo-inverse	$\mathbb{R}^{(N-1)\times16}$	–
A	learned state matrix	$\mathbb{R}^{12\times12}$	–
B	learned input matrix	$\mathbb{R}^{12\times4}$	mixed

State estimation (block 11)

$\hat{\boldsymbol{x}}_k$	EKF estimate of $\boldsymbol{x}_k$	$\mathbb{R}^{12}$	as $\boldsymbol{x}_k$
Σ_k	estimate covariance (how unsure)	$\mathbb{R}^{12\times12}$	m^2, rad^2, …
Q_e	process noise, from DMDc residuals	$\mathbb{R}^{12\times12}$	as Σ_k
$\boldsymbol{\ell}_j$	known floor marker j (world)	$\mathbb{R}^3$	m
$\boldsymbol{q}_j$	marker j in camera axes	$\mathbb{R}^3$	m
$\boldsymbol{z}_j$, $\boldsymbol{z}_\omega$	pixel of marker j, gyro reading	$\mathbb{R}^2$, $\mathbb{R}^3$	px, rad/s
f, c_x, c_y	focal length, image centre	$\mathbb{R}$	px (400, 320, 240)
R_{bc}	camera axes in the body frame	$\text{SO}(3)$	–
$h(\cdot)$, H	measurement model, its Jacobian	$\mathbb{R}^{12} \to \mathbb{R}^m$, $\mathbb{R}^{m\times12}$	px
R_n	measurement-noise covariance	$\mathbb{R}^{m\times m}$	px^2, $(\text{rad/s})^2$
L	Kalman (estimator) gain	$\mathbb{R}^{12\times m}$	mixed

Optimal control (blocks 3–6)

J	total cost of the flight	$\mathbb{R}$	–
Q	state weight, $Q \succeq 0$	$\mathbb{R}^{12\times12}$	$1/(\text{state unit})^2$
R	input weight, $R \succ 0$ (not the attitude)	$\mathbb{R}^{4\times4}$	$1/\text{N}^2$, $1/(\text{N m})^2$
$V(\boldsymbol{x})$	value function (lowest cost from $\boldsymbol{x}$)	$\mathbb{R}$	–
P, P_k	value matrix (iteration k)	$\mathbb{R}^{12\times12}$	as Q
K	LQR feedback gain	$\mathbb{R}^{4\times12}$	mixed

MPC and QP (blocks 7–9)

N (blocks 7–9)	prediction horizon	$\mathbb{N}$	steps (25)
P_f	terminal weight (= P from DARE)	$\mathbb{R}^{12\times12}$	as Q
$\mathcal{U}$, $\mathcal{X}$	admissible inputs / states	sets	–
$\mathbf{U}$	stacked inputs $[\boldsymbol{u}_0; \ldots; \boldsymbol{u}_{N-1}]$	$\mathbb{R}^{4N}$	N, N m
S_x, S_u	prediction matrices	$\mathbb{R}^{12N\times12}$, $\mathbb{R}^{12N\times4N}$	–
$\bar{Q}$, $\bar{R}$	block-diagonal weights	$\mathbb{R}^{12N\times12N}$, $\mathbb{R}^{4N\times4N}$	–
H, $\boldsymbol{f}$	QP Hessian, linear term	$\mathbb{R}^{4N\times4N}$, $\mathbb{R}^{4N}$	–
G, $\boldsymbol{h}$	inequality constraints	$\mathbb{R}^{n_c\times4N}$, $\mathbb{R}^{n_c}$	mixed
A_{eq}, $\boldsymbol{b}_{eq}$	equality constraints	$\mathbb{R}^{n_e\times4N}$, $\mathbb{R}^{n_e}$	mixed
$\boldsymbol{u}_0^*$	first optimal input (applied)	$\mathbb{R}^4$	N, N m

Actuation (block 10)

$\boldsymbol{u}_{\text{trim}}$	hover input $[mg_0; 0; 0; 0]$	$\mathbb{R}^4$	N, N m
Γ, $\boldsymbol{f}$	mixer matrix, rotor thrusts	$\mathbb{R}^{4\times4}$, $\mathbb{R}^4$	–/m, N
k_f, $\boldsymbol{\Omega}$	thrust constant, rotor speeds	$\mathbb{R}$, $\mathbb{R}^4$	N s^2, rad/s